

Key Takeaways

1 IPCC sea-level projections are biased low by documented, quantifiable amounts

1.1 Literature context

IPCC AR6 medium-confidence projections for 2100 range from ~ 0.3 m (SSP1-2.6) to ~ 0.7 m (SSP5-8.5) relative to a 1995–2014 baseline (*Fox-Kemper et al.*, 2021). These estimates are built on ISMIP6 ice sheet models (*Seroussi et al.*, 2020), GlacierMIP emulators (*Marzeion et al.*, 2020), and coupled climate model output for thermal expansion.

Multiple independent lines of evidence show that these projections systematically underestimate sea-level rise. ISMIP6 ice sheet models diverge from IMBIE satellite observations almost immediately, with most models producing ice sheets that gain mass under warming rather than losing it (*Aschwanden et al.*, 2021; *Fricker et al.*, 2025). Observed trends in global mean sea level under continued warming exceed IPCC medium-confidence projections for all emission scenarios (*Hamlington et al.*, 2025). The universal assumption of Glen’s flow law exponent $n = 3$, when satellite and laboratory observations support $n \approx 4$, produces a 21–35% underestimation of 100-year sea-level rise (*Millstein et al.*, 2022; *Martin et al.*, 2026; *Getraer and Morlighem*, 2025; *Fan et al.*, 2025). Calving, subglacial hydrology, and ice damage are absent from most models (*Aschwanden et al.*, 2021; *Dow et al.*, 2022), and sensitivity to calibration methodology alone produces factor-of-two differences in projected loss from the same model (*Goldberg et al.*, 2026). Parametric uncertainty, which is omitted from standard intercomparisons, may exceed inter-model spread (*Aschwanden et al.*, 2019).

1.2 Unique contribution

Prior work has identified individual sources of low bias in isolation: rheology (*Millstein et al.*, 2022; *Martin et al.*, 2026), missing physics (*Aschwanden et al.*, 2021), calibration sensitivity (*Goldberg et al.*, 2026), and observational exceedance (*Hamlington et al.*, 2025). However, no study has constructed a complete, observation-calibrated alternative framework that avoids these biases while producing conditional predictive distributions with propagated uncertainty.

This work provides that alternative. By calibrating exclusively against observational data using physics-informed Bayesian models at the component level, the framework is structurally independent of the process models that produce the low bias. Each component model makes its calibration range, functional form, and extrapolation assumptions explicit. The result is not merely a critique of IPCC projections but a quantitative replacement that identifies where and by how much the process-model estimates depart from what the observations support. For glaciers, the data-driven temperature sensitivity ($b_{gl} = 0.77 \text{ mm yr}^{-1} \text{ }^{\circ}\text{C}^{-1}$) is approximately $6\times$ the value implied by IPCC medians ($b_{IPCC} \approx 0.13 \text{ mm yr}^{-1} \text{ }^{\circ}\text{C}^{-1}$), and the IPCC rate structure is inverted: attributing most mass loss to a committed baseline rather than to ongoing warming. For Greenland discharge, physical derivations of all model parameters from glacier dynamics agree with fitted values to

within observational uncertainty, providing independent validation that the data-driven framework captures the governing physics.

2 West Antarctic uncertainty dominates the projection envelope and is wider than current assessments acknowledge

2.1 Literature context

The West Antarctic Ice Sheet (WAIS) contains ~ 3.3 m of sea-level equivalent and is grounded below sea level on retrograde beds, making it susceptible to marine ice sheet instability (MISI). Grounding-line retreat is observed at Thwaites Glacier (*Joughin et al.*, 2014), and ocean thermal forcing sufficient to drive WAIS retreat is committed regardless of emissions pathway (*Naughten et al.*, 2023). Present-day forcing held constant may be sufficient to deglaciate large parts of WAIS (*van den Akker et al.*, 2025).

Robel et al. (2019) showed analytically that MISI amplifies and skews uncertainty toward worse outcomes: when the grounding-line flux nonlinearity exponent exceeds 3 (realistic values are 4–5), ensemble distributions are systematically right-skewed. *Bamber et al.* (2019) elicited WAIS-specific probability distributions from 22 ice sheet experts, finding a 5th–95th percentile range of -50 to 930 mm under $+5^\circ\text{C}$ warming—far wider than ISMIP6 ranges. Evidence against marine ice cliff instability (MICI) as a 21st-century mechanism has grown (*Morlighem et al.*, 2024; *Clerc et al.*, 2019; *Schlemm et al.*, 2022), but MISI itself, combined with processes that current models omit (calving, subglacial hydrology, incorrect rheology), produces substantial uncertainty without invoking MICI.

2.2 Unique contribution

Existing treatments of WAIS uncertainty either rely on process models that omit the dominant physics (*Seroussi et al.*, 2020) or provide expert judgment distributions without temporal structure (*Bamber et al.*, 2019). No prior work has combined (i) a scenario-mixture framework with physically motivated within-scenario skewness derived from MISI dynamics (*Robel et al.*, 2019), (ii) a trajectory exponent that distributes mass loss within the century based on grounding-line flux scaling (?), and (iii) a post-hoc rheology correction reflecting the $n = 3 \rightarrow 4$ bias (*Martin et al.*, 2026; *Getraer and Morlighem*, 2025).

This work constructs a three-scenario mixture (status quo, MISI, MISI+MICI) with scenario weights, skewness parameters, and trajectory exponents each grounded in specific physical mechanisms. The framework produces a 90% credible interval of approximately 60–1800 mm by 2100, wider than the *Bamber et al.* (2019) expert judgment (which predates the rheology bias quantification) but in broad distributional agreement, providing independent methodological corroboration. The temporal structure—back-loaded mass loss with $\sim 30\%$ of the 2100 value realized by 2050—is consistent with expert expectations (*Bamber et al.*, 2019) and the grounding-line flux physics (*Robel et al.*, 2019), but is absent from existing probabilistic frameworks. By making the scenario weights, physical justifications, and correction factors explicit and modifiable, the framework is transparent and updatable as new observations and process understanding emerge.

3 The observed acceleration in sea-level rise originates from specific, identifiable component-level sources

3.1 Literature context

Global mean sea level is accelerating, rising at $\sim 4.5 \text{ mm yr}^{-1}$ in the most recent decade compared to $\sim 1.4 \text{ mm yr}^{-1}$ over the 20th century (*Frederikse et al.*, 2020; *Hamlington et al.*, 2020). The budget reconstruction of *Frederikse et al.* (2020) closed the 20th-century sea-level budget by attributing contributions to thermal expansion, glaciers, ice sheets, and land water storage. However, identifying the temperature sensitivities of individual components—and thus the physical drivers of the acceleration—requires fitting rate–temperature relationships to each contributor separately.

For Greenland, *King et al.* (2020) showed that discharge acceleration was driven by calving-front retreat with a consistent 4–5% speedup per km of retreat. *Feliksion et al.* (2017) quantified how dynamic thinning propagates inland, setting the timescale for the ice sheet response. Subsurface ocean warming drives submarine melt and terminus retreat (*Straneo et al.*, 2013; *Holland et al.*, 2008; *Xu et al.*, 2013; *Rignot et al.*, 2016; *Jackson et al.*, 2022), but the aggregate relationship between ocean temperature and ice-sheet-wide discharge had not been formalized as a calibrated predictive model. For glaciers, the GlAMBIE consensus (*Hock et al.*, 2024) provides the first globally complete, observation-only mass-balance record, but no prior study has extracted a global temperature sensitivity from it and compared that sensitivity against process-model predictions (*Rounce et al.*, 2023) and IPCC projections.

3.2 Unique contribution

This work provides the first unified component-level decomposition in which every contributor’s temperature sensitivity is extracted from observations through a common Bayesian framework, enabling direct comparison of sensitivities across components and against IPCC process-model values.

For Greenland discharge, the time-delay model ($R^2 = 0.995$) identifies a 5–6 year lag between subsurface ocean warming and the discharge response, with the coupling $\gamma = 0.365 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-1}$ independently derivable from glacier dynamics—a chain linking submarine melt sensitivity (*Xu et al.*, 2013; *Rignot et al.*, 2016), calving amplification (*Benn et al.*, 2007; *Luckman et al.*, 2015), and velocity response to retreat (*King et al.*, 2020). The agreement between three independent estimates (observational, bottom-up, and fitted; all within $\pm 10\%$) provides confidence that the model captures the governing physics rather than fitting noise. No prior study has closed this parameter-estimation loop for aggregate Greenland discharge.

For glaciers, the data-driven sensitivity $b_{gl} = 0.77 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-1}$ from GlAMBIE is approximately $1.5\text{--}2\times$ the effective sensitivity of the PyGEM process model (*Rounce et al.*, 2023) and $\sim 6\times$ the value embedded in IPCC medians. This provides an observational benchmark that process models must reproduce to be considered validated at the global aggregate scale—a test that has not previously been applied.

4 Simpler, observationally anchored models are better suited to the forecasting task than complex process models

4.1 Literature context

The forecasting literature, developed across economics, meteorology, epidemiology, and supply chain management, has established that increasing model complexity beyond the information content of the calibration data degrades out-of-sample skill (*Green and Armstrong*, 2015). Forecast combinations outperform individual methods (*Makridakis et al.*, 2020). Model skill must be demonstrated through out-of-sample validation, not in-sample fit.

Current sea-level projections rely on process models with millions of degrees of freedom constrained by observational records spanning at most a few decades of the relevant forcing range. IS-MIP6 models are initialized to the present state and run forward without systematic out-of-sample validation against withheld data (*Aschwanden et al.*, 2021; *Fricker et al.*, 2025). The sensitivity of projections to initialization, spin-up, and calibration methodology (*Goldberg et al.*, 2026) suggests that these models operate in the regime where complexity degrades rather than enhances forecast skill.

4.2 Unique contribution

This work applies established forecasting principles to sea-level prediction for the first time as a systematic framework. The hierarchy of increasing complexity—naive extrapolation, aggregate semi-empirical, component-level physics-informed—follows the forecasting principle that complexity should be added only when it demonstrably improves out-of-sample skill.

Each component model has ~ 3 – 6 free parameters calibrated against ~ 25 – 70 independent observations, a ratio of data to parameters that the forecasting literature identifies as conducive to reliable out-of-sample prediction. By contrast, ISMIP6 models have millions of degrees of freedom constrained by comparable observational records. The framework validates against independent datasets withheld from calibration: *Mankoff et al.* (2021) and NOAA Arctic Report Card for Greenland discharge, *Frederikse et al.* (2020) and *Horwath et al.* (2022) for glaciers and thermosteric, and IMBIE (*Otosaka et al.*, 2023) for total Greenland mass balance.

The structural difference from process models is not merely a matter of parsimony. Each component model makes its calibration range and extrapolation lever arm explicit: the glacier model is calibrated over $\sim 1^\circ\text{C}$ of warming, and predictions at $+3$ – 4°C are identified as extrapolations with corresponding structural risk. Process models embed analogous extrapolation assumptions in their parameterizations without reporting them. The learned model-inadequacy parameter σ_{extra} in each component quantifies the systematic misfit within the calibration range, providing a transparent lower bound on structural uncertainty that process models do not report.

5 Hierarchical coupling across components propagates uncertainty faithfully and identifies where observations would most reduce it

5.1 Literature context

Sea-level projection uncertainty is typically reported as a total range without decomposing contributions from individual components or identifying which observations would most reduce that range. IPCC AR6 reports separate component projections (*Fox-Kemper et al.*, 2021), but these are produced by different modeling groups using different assumptions, and the uncertainty propagation from forcing scenarios through component models to the total is not performed within a unified statistical framework.

The thermosteric component involves a well-understood physical lag: the ocean absorbs heat on a timescale $\tau_u \sim 5$ – 25 yr (?), creating committed sea-level rise from warming already absorbed by the atmosphere but not yet expressed in thermal expansion. Greenland discharge responds to subsurface ocean temperature (*Straneo et al.*, 2013; *Holland et al.*, 2008), creating a physical coupling between the ocean thermal state and ice sheet dynamics. These two components share a common physical driver (ocean heat content) but have been modeled independently in all prior projection frameworks.

5.2 Unique contribution

This work introduces a shared latent ocean state that couples the thermosteric and Greenland components within a single Bayesian inference. The thermosteric model’s ODE state $S_u(t)$ is jointly calibrated against NOAA steric sea level (?) and EN4 subsurface temperature (*Good et al.*, 2013) through a dual likelihood that resolves the τ_u - b identifiability problem present in single-likelihood fits. The jointly estimated transfer function $T_{\text{sub}} = \kappa S_u + \delta$ then provides the ocean temperature forcing for the Greenland discharge model, so that uncertainty in the ocean thermal state propagates consistently into both the thermosteric and Greenland discharge projections.

This coupling has two consequences absent from existing frameworks. First, scenarios that produce greater committed thermosteric rise (larger τ_u , implying a slower ocean that has absorbed less of the surface warming signal) simultaneously produce warmer subsurface temperatures that drive greater Greenland discharge—a positive correlation between components that widens the total projection envelope relative to what independent-component summation would produce. Second, the decomposed uncertainty budget identifies WAIS dynamics as the dominant source of total projection uncertainty, with Greenland discharge sensitivity and the ocean thermal lag as secondary contributors. This ranking is actionable: it identifies where additional observations (e.g., longer subsurface ocean temperature records, continued GRACE mass-balance monitoring) would most efficiently reduce projection uncertainty, and where uncertainty is irreducible on policy-relevant timescales (WAIS instability dynamics).

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